

Open replicators evolve closure in a digital primordial soup

Draft assembled 2026-10-08 from `manuscript/MAIN_nature.md` and `manuscript/figures/out`; author list and affiliations to be added.

Nature Article draft v2, 2026-10-08. Title 61 characters. Summary ≤ 200 words. Subheadings ≤ 40 characters. Main text target $\leq 3,500$ words. Every number is from a generated table named in the Methods (Data availability) and in `manuscript/NUMBERS_INDEX.md`.

Theories of the origin of life disagree about what came first but agree that at some point a self-propagating pattern became an individual, something whose persistence depends on itself rather than on its surroundings¹⁻⁴. That step has never been watched. Soups of random computer programs on a shared memory produce self-replicators without any fitness function⁵⁻⁷, so the step can be observed, repeated and taken apart. Here we show, in thousands of worlds of random Z80 machine code, that heredity precedes individuality. The first replicator is a two-byte literal that writes itself, open to its surroundings and dependent on them; closure, a control-flow cycle that keeps the organism's execution inside its own bytes, evolves afterwards under the damage the open form suffers, and independent worlds converge on byte-identical closed genomes. We prove that closure requires such a cycle and that a machine's smallest replicator is open exactly when its instruction set can write a literal. The Brainfuck-family soup of Agüera y Arcas et al.⁵ is born closed as predicted; one added literal-writing instruction makes it start open and shows that the open phase is a window whose length the tar decides. Individuality here is a property of a program's control, not a boundary in space.

Main

Life is often defined by a list — metabolism, compartment, heredity — and origin-of-life research by which item came first⁸⁻¹¹. A different tradition asks what makes a living thing an *individual*: Maturana and Varela's operational closure¹, Rosen's closure to efficient causation², Fontana and Buss's organizational closure³, and the information-theoretic individuality of Krakauer and colleagues⁴ all locate life in a system's dependence on itself rather than on its environment. Neither tradition has had an experimental system in which the transition from pattern to individual can be watched, replicated and dissected.

Soups of random programs offer one. Agüera y Arcas et al. showed that random Brainfuck-like programs executing on a shared tape give rise to self-replicators without any fitness function⁵; the Coreworld⁶ and random Lisp programs⁷ showed the same earlier, and random Z80 programs have recently been used to study the coevolution of replication and function¹². These systems have nothing to protect — no energy, no membrane, no explicit selection — so whatever becomes an individual there does so by the logic of self-reference alone. We built such a soup from a real instruction set, the Zilog Z80, on the GPU, with an assay that measures heredity by intervention rather than inspection, and ran it as an experiment: thirteen instruction-set ablations, tape lengths from 3 to 100 bytes, execution budgets from 32 to 512 steps per encounter, mutation rates over two decades, and pre-registered confirmatory stages with kill criteria.

A soup with nothing to select for

The soup holds 20,000 programs ("tapes") of L bytes, randomly initialised (Fig. 1a). At each step 8,192 random cells are drawn, each paired with one of its four neighbours on a 160×125 lattice, and about 3,980 of these pairs are active; each pair is concatenated into a 2 L -byte ring, executed by a Z80 core for 128 instructions starting at the first byte of the first tape with the stack pointer at the last byte of the second, and written back. A background mutation replaces a fixed number of random bytes per step. There is no fitness function and no replacement: a tape changes only by being executed or written into. The sweeps ran on 50 parallel GPUs (Methods).

Replication is measured by a *culture test* (Fig. 1b): a tape is executed against 64 random partners; the fraction of partners that become $\geq 75\%$ copies of it is its copy success, and the tape is *heritable* if its offspring, run against fresh partners, produce copies in turn ($\text{gen2} \geq 0.3$). The test matters because the most common byte pattern in a soup is often not a replicator: a program may write an inert pattern into its neighbours, and a return-address smear can occupy 0.5% of the soup without inheriting anything. We date emergence by the first heritable exemplar among the three most common tapes with a share of at least 0.5%, and follow the heritable fraction of random cells as the population measure.

Tar, then an open replicator

In every unablated world that produces life the same sequence follows (Fig. 1c,d). First comes *tar*: at the moment the first pattern reaches 10% occupancy, 24–34% of all bytes are zero at every tape length from 9 to 100. The zeros are self-written — `PUSH`, `CALL` and `RST` with zero registers and return addresses write zeros wherever the stack pointer points, and zero is a no-op — compressible, spreading and sterile: its carriers have $\text{gen2} \approx 0$. It is the digital analogue of Benner’s *tar*¹³, order without heredity made by the same chemistry that will make life.

Then the first heritable replicator appears, and it is the same program almost everywhere: the two-byte word `01 c5, LD BC, nn ; PUSH BC`, tiled across the tape. In the pre-registered confirmatory stage it is the modal first tape in 19/20, 12/20, 14/20 and 18/20 worlds at $L = 16, 20, 50$ and 64 , the rest being its register variants, at a Kaplan–Meier median of 450–750 steps regardless of L (Fig. 1d). Its mechanism is a tautology (Fig. 1b): the literal `01 c5` is both the instruction `LD BC, $01c5` and the data it loads; `PUSH BC` writes those two bytes onto the stack, which lies in the partner, and the pointer advances to the next copy of the same literal. Code equals data equals literal. It needs no loop, because the ring brings the pointer back; no counter, because every byte of the tape is the same word; and no test of its surroundings, because it never reads them. It is also nearly the only thing of its size that can replicate: of all 65,536 two-byte words tiled to 16 bytes, exactly five leave an all-zero partner holding a copy, all load–push pairs (`01 c5, 11 d5, 21 e5`, and the memory-load variants `2a e5/e5 2a`), and none copies into more than 81% of random partners. A further 246 words pass the heredity threshold without producing a single 75% copy — `CALL` instructions smearing return addresses — which is why we require faithfulness as well as heredity before calling anything a replicator.

Closure evolves without a membrane

The pusher does not stay. It is *open*: executed against 256 random partners it copies into a median 0.66–0.73 of them (range 0.56–0.84 over 80 worlds and four lengths) and loses a quarter or more of its own bytes in a median 0.06–0.43 of encounters (Fig. 2a,b). The failures have one cause. After writing its copy the pointer runs on into the partner and executes the partner’s bytes, which can poison the register pair before the next push. Against all-zero partners the pusher copies perfectly; its success falls as the density of active bytes in the partner rises, and no execution budget rescues it.

By 300,000 steps the dominant tape at $L = 16$ carries a control-flow instruction in 20/20 worlds and 17 of them end on the identical sequence `ad e3 21 e3 21 c0 ad c0 — XOR L ; EX (SP), HL ; LD HL, nn ; RET NZ` — whose `RET NZ` returns the pointer into the organism’s own body; at $L = 50$ all 20 worlds carry a `JR NZ` inserted into the pusher tiling, 14 of them byte-identical (Fig. 2e). Across the 233 unablated worlds of the exploratory stages, 92 gained a control-flow instruction between the first replicator and the final dominant and 3 lost one (exact McNemar $P = 7 \times 10^{-24}$). Every such successor copies into 1.00 of 256 random partners with 0.00 self-damage, whatever instruction implements the cycle: `RET NZ`, `JR NZ`, `DJNZ`, `JP, JP (HL), RET`, or the hardware loops `LDIR` and `LDDR` (Fig. 2a,b). The population tells the same story (Fig. 2c): at $L = 16$ the heritable fraction of random cells stays between 0.07 and 0.19 for the first 20,000 steps after emergence — the pusher’s reign is a world of damaged copies — and rises to 0.75–0.94 when the closed successor takes over.

We call this property *closure*: the organism’s bytes are a forward-invariant set under execution for every context (Methods). It is Maturana and Varela’s operational closure made countable, and Krakauer’s individuality made measurable by intervention: the closed organism’s future depends on its own past alone. Note what closure is not. It is not a membrane; the closed organism’s bytes are as exposed to writes as before. It is not a test of the environment; `RET NZ` tests a flag the organism set itself. It is a cycle in the organism’s control flow that never leaves the organism (Fig. 2d). Closure takes longer the larger the organism: at $L = 20$, 19/20 worlds are closed by one million steps, at $L = 64$ only 8/20, with 12 still dominated by the open pusher, which there is also the least damaged open form in the atlas (self-damage 0.06) and so under the weakest pressure to close. The pre-registered expectation of $\geq 10/20$ closures at both lengths was met at $L = 20$ only behaviourally — that family closes with `LDIR`, which our syntactic criterion did not count — and not at $L = 64$ (Methods, Pre-registration).

Three load-bearing instructions

Which instructions does this need? The seed-paired ablation atlas (Fig. 3a) finds three load-bearing parts and nothing else. Removing *immediate loads* leaves 2/10 worlds alive at 300,000 steps; removing *stack writers* delays emergence 160- to 992-fold in median and hands the soup to a replicator built on the block-copy instruction `LDIR`; removing *all copying* (loads, pushes and block copy, 112 opcodes) or leaving only read-modify-write instructions produces no replicator in 40

worlds at one million steps. Every other ablation — block copy, calls and returns, memory loads, I/O, exchanges — is within seed-paired noise of the unablated soup. The mutation rate chooses the family that holds the soup, and the execution budget decides whether an organism can be faithful: at $L = 100$ the `LDIR` regime is faithful in 1/10 worlds at 128 steps per encounter and 10/10 at 256, while at 64 steps no exemplar ever reaches 0.5% of the soup yet 6/10 final soups are heritable populations with no clone — heredity carried by partial copies, the communal regime Woese placed before the Darwinian threshold¹⁴.

At $L = 9$ removing the stack writers makes life *more* likely (12/20 vs 6/20 worlds at 128 steps and 19/20 vs 5/20 at 512, Cochran–Mantel–Haenszel $P = 3.9 \times 10^{-6}$; Fig. 3d), because the tar they produce destroys nascent `LDIR` copiers before they assemble: an `LDIR` tape reaches the ten most common tapes in 5/20 worlds at a median 118,500 steps with stack writers present, against 20/20 within 3,400 steps without, and programs are destroyed 6.8 against 1.3 times per 1,000 interactions. Tar is not a nursery; it is a hazard that only an open literal-writer, which cannot be poisoned by zeros, gets through.

No size floor, a geometry switch

Emergence is certain from $L = 16$ to $L = 100$ (10/10 worlds at every length under four control arms) and its median time rises only eightfold over that range, and not at all when the execution budget grows with the tape (Fig. 1d): the pusher costs two bytes however long the tape, because tiling makes a large organism of a small word. Below 16 there is no monotone floor but a dead zone (Fig. 3b): life at $L = 5$ and 6 (whole-tape and period-3 copiers, slowly), at $L = 8$ and from $L = 16$ up; none at $L = 3, 4$ and 7; and 4/10, 4/10 and 6/10 worlds at $L = 9, 10$ and 12, exactly where the pusher’s own isolated heritability dips to 0.20–0.32. The dead zone is geometric: the stack pointer descends two bytes per push while the instruction pointer ascends, and on a $2L$ -byte ring they race; when a push lands on the literal the pointer is about to read, the register pair is poisoned. Padding the pair memory to a longer ring without changing the organism moves the collision and switches the dead zone off (Fig. 3c): at $L = 12$, rings of 28 or 36 bytes give 10/10 worlds alive at 350–600 steps where the native 24-byte ring gives 6/10 at a median 245,500. Closure, too, has a geometric channel: the `RET NZ` closer returns to an absolute address and copies 0.00 of partners on any ring but its own, whereas a relative-jump closer keeps control closed on every ring. Control can be made coordinate-free; this machine’s reproductive channel cannot.

Tar fools compression detectors

The transition of Agüera y Arcas et al. is detected by a drop in high-order entropy, byte entropy minus compressed size⁵. In our unablated soups that detector is anti-correlated with life at $L = 25, 49, 64$ and 81 (area under the curve 0.07–0.34 against the culture-test ground truth), because the zero flood compresses better than the replicators that replace it; the fraction of non-unique tapes and the raw bits per byte are robust (≥ 0.84 at every length but one), and occupancy of the dominant genotype is a poor proxy for a heritable population (0.61–0.95 by length). Tar is more orderly than life, and a one-byte organism, as we show below, is as compressible as tar. Any agnostic biosignature that scores order without testing propagation will be fooled by the chemistry that makes life possible.

A literal channel starts life open

Why is the first replicator open, and must it be? Two counting arguments answer (Methods, Theorems). A straight-line program’s pointer runs on into its partner, so a straight-line replicator is open by construction; and a closed replicator must execute more bytes of code than it has, because each byte of code writes fewer than one byte, so its pointer must revisit an address: closure requires a cycle. A straight-line replicator exists only if the instruction set can write a *literal* — content independent of the memory being overwritten — at a self-advancing pointer. The Z80 can (`LD rr, nn ; PUSH rr`). BFF, the Brainfuck family of Agüera y Arcas et al., cannot: `+` and `-` increment in place, `.` and `,` copy from memory, and a one-pass execution of 128 instructions cannot copy a 64-byte tape with three or more distinct bytes (it needs 64 writes, 63 head moves and 2 source changes). An exhaustive search over 16.4 million structured straight-line BFF programs under a wrapping pointer found only one-symbol fills, the BFF analogue of tar.

Eighty-four soups of 2¹⁷ programs run for 16,384 epochs test the consequences (Fig. 4). In BFF as published, 9 of 24 worlds produced a heritable replicator; every first replicator contains a loop (9/9), none is open (0/9), and seven of nine are closed by our strict criterion from the first sample in which they appear. A wrapping pointer produces more transitions (19/24; Fisher $P = 0.008$) but the same birth: 19/19 loop-bearing, 0/19 open, 16/19 closed. Adding one instruction that writes its own two operand bytes below the write head — the literal channel — changes the beginning completely: in 12/12 worlds the first replicator is the one-byte tiling of that instruction, whose operands are two more copies of itself, open (its

pointer enters the partner in 100% of encounters), loop-free, and present at the first sample, epoch 64, with 53–60% of the soup. It is the 01 c5 of BFF. And then it dies: within about 200 epochs each of the 12 soups turns into tar that halts a pointer within a few dozen steps (an unmatched] ends BFF execution), copies into such partners are cut short and sterile, and the open replicator persists only as a 0.2% minority, heritable in isolation and unable to spread. No closed class evolves, and none can: no closed heritable replicator exists among 1.3 million periodic programs of this language up to period 10 (Methods). The open phase is therefore a *window*, not a stage, and one more switch shows what sets its length. Making an unmatched bracket a no-op instead of a halt — benign tar — leaves the same open replicator in place in 12/12 worlds for all 16,384 epochs, with every random tape heritable at the end and no collapse; and since no closed design exists in this language (none among 1.3 million periodic programs up to period 10, Methods), none evolves: a substrate in which life is open forever. Keeping the literal but removing the wrapping pointer still starts life open in 12/12 worlds, but the one-pass bandwidth makes the wave weak (it reaches half the soup in 5/12 worlds) and it dies out in all. Together these complete a classification (Fig. 4d): a self-consistent literal channel decides whether life starts open or closed; the lethality of the tar decides whether an open beginning lasts; and the instruction set’s supply of copyable closers decides whether a lasting open phase ends in closure, as in the Z80, or never does.

Closure as a theorem

The architectural statement holds for any sequential shared-memory machine whose pointer runs on into adjacent memory by default and whose code writes less than one byte per byte executed (Methods, Theorem 6): every closed replicator contains an executed cycle; a self-consistent literal-write instruction yields a straight-line, hence open, replicator the size of the literal; and so the smallest open replicator is strictly smaller than the smallest closed one and is present in a random soup with exponentially higher probability. In information terms (Lemma 7) closure is zero inflow of information from context to offspring, $I(o; e | x) = 0$, the limit that perfect fidelity across contexts requires; selection for fidelity is selection against listening to the environment. What the theorems do not decide is the dynamics after the open beginning: closure arrives with probability $\approx 1 - \exp(-q \int n dt)$, where $\int n dt$ is the integral of the open population over time and q the rate of closing mutations. Benign tar (Z80 zeros are no-ops) lets the integral grow until closure is found; lethal tar (BFF brackets halt the pointer) closes the window first (Fig. 5).

Discussion

Three claims rest on these results. First, in soups of random machine code heredity precedes individuality: the first replicator is a self-writing literal that is open to its surroundings and depends on them, and closure is an evolved refinement selected by the damage the open form suffers. Second, closure is topological rather than geometric: a cycle in control flow, not a boundary in space, reached by different instructions in different worlds that converge on the same function and often the same bytes. Third, the elements a substrate must offer are few and specific — a literal that is its own code, a write channel aimed at a neighbour, and a cycle — and the stages they produce are the ones origin-of-life theory has argued over in the abstract: Benner’s tar¹³, Eigen’s error threshold^{15,16} (which we observe as a curve in the L_{DIR} regime), Woese’s communal heredity before vertical descent¹⁴, the prelife of Nowak and Ohtsuki¹⁷, and Fontana and Buss’s organizational closure³, here arising from a copier rather than in its absence.

Whether anything of this transfers to chemistry is open. What the system offers is a definition of closure that can be checked by intervention — execute the candidate in many contexts and ask whether its own bytes survive — and a warning for life detection: the first ordered, spreading, compressible thing in these worlds is not alive, and the first living thing is a two-byte tautology that no inspection of structure would flag^{18,19}.

Methods

Soup. The soup holds $N = 20,000$ tapes of L bytes (L from 3 to 100), initialised with uniformly random bytes, on a 160×125 square lattice with local interaction. At every step 8,192 cells are drawn uniformly at random and each is paired with one of its four lattice neighbours (edges clamped, no wrap-around; a cell paired with itself is inactive); a pair is active if neither member is already in another pair that step, which leaves $\approx 3,980$ active pairs (0.395 active interactions per cell per step; the count varies by a few per cent with concurrent GPU load and does not affect results). An active pair is concatenated, organism first, into a pair memory of P bytes ($P = 2L$ except in the padding experiment of Stage F, where $P \geq 2L$ and the padding bytes are never written back), executed by the Zilion Z80 core (@neovand/zilion 0.2.0, WGS�)

for a fixed budget of 128 instructions (32, 64, 256, 512, 1,024 or 2,048 in the budget arms) from byte 0 with all registers zero and the stack pointer at the last byte of the partner, and both tapes are written back. After each step $8,192/2^k$ bytes chosen uniformly at random over the soup are replaced by random bytes ($k = 4$ unless stated; $k = 1-6$ in the mutation arms). There is no fitness function, no replacement and no death. Instruction-set ablations replace whole opcode families by no-ops through a prefix-aware fetch hook that sees the operation after DD/FD and ED/CB prefix resolution; the thirteen ablations and their opcode counts are listed in Extended Data Table 2 (none, block-copy (4 opcodes), stack-writes (46), stack-write-only (22), stack-read-only (24), push-only, ex-sp-only, call-rst, call-rst-write, ld-mem, ld-imm, ld-reg, all-ld, ed-loads, cb-page, no-copy (112), rmw-only). The emulator was validated against an independent Z80 reference with zero genuine divergences in randomised 5,000-pair tests with and without ablation; it is not treated as ground truth for the Z80 itself, and no claim depends on cycle-exact timing.

Recording. Every run writes a sample at fixed steps (every step for the first 30, then a geometric schedule, then every 500 or 1,000) with: the ten most common exact tapes and their shares; quasispecies occupancy (fraction of cells within Hamming distance $\lceil L/4 \rceil$ of the dominant tape); byte histogram, zero-byte fraction and byte entropy; brotli-compressed bits per byte (quality 2) and the high-order entropy of ref. 5; species entropy and unique-tape fraction; per-interaction statistics (bytes written into the partner, copy events, destroyed programs); 16 random tapes for the functional census; and full soup snapshots at fixed steps and at emergence.

Culture test and partner tests. A tape runs as the first program against 64 uniformly random partners for the condition's budget; copy success is the fraction of partners that become $\geq 75\%$ copies at the best cyclic shift; gen2 is the copy success of the produced copies against fresh partners, weighted by the copy fraction; heritable means $\text{gen2} \geq 0.3$, faithful additionally $\geq 50\%$ of partners copied. Partner-independence tests use 256 random partners and record self-damage ($\geq 25\%$ of the organism's bytes changed in one encounter). In BFF the culture test also records whether the instruction pointer entered the partner in each encounter; closed means $\leq 5\%$ of encounters entered and $\geq 95\%$ of partners copied, open means $\geq 50\%$ entered.

Events and statistics. t_{rep} is the first sampled step at which one of the three most common tapes has a share $\geq 0.5\%$ and is heritable by the culture test (a clonality criterion by design: it dates the first *clone*, not the first heritable material); t_{faith} additionally requires faithfulness; $t_{q_{10}}$ (quasispecies occupancy $\geq 10\%$) is the pre-registered occupancy event and is reported because it fires on zero floods as well as on replicators. Emergence fractions are quoted as worlds reaching the event within the horizon with Wilson 95% intervals; times as Kaplan–Meier medians with censoring at the horizon. Ablation effects are ratios of Kaplan–Meier medians with seed-paired sign tests (the same seeds run under every ablation); confirmatory contrasts use one-sided Fisher exact tests and the Cochran–Mantel–Haenszel test pooled over budgets; the paired gain of control flow uses the exact McNemar test; between-variant transition rates in BFF use two-sided Fisher tests and Mann–Whitney tests on event times with non-transitioning runs censored at the horizon. In BFF, where replicator populations are quasispecies with small exact-class shares, the "first replicator" is the most common tape class (a tape and its reverse count as one class) at the first sample at which it is heritable, and the population event is a heritable fraction ≥ 0.5 of 32 random tapes; t_{rep} is reported alongside. All thresholds (0.75 copy, $\text{gen2} \geq 0.3$, 0.5% share, 10% occupancy, 95%/5% closure) were fixed before the confirmatory stages. Same-seed Z80 runs diverge because GPU scheduling makes the pairing order non-deterministic, so every Z80 claim is a population claim over seeds (ten same-seed repeats spread as widely as ten seeds); the BFF soup is bitwise reproducible across machines (identical epoch statistics on an Apple GPU and an NVIDIA L40S for six seeds over 16,384 epochs).

Definitions and theorems. A *self-writer* is a tape fixed under its own execution in some context. *Closure*: the organism's bytes are forward-invariant under execution for every context in the class of uniformly random partners. Theorem 1: no straight-line execution in one-pass BFF copies a 64-byte tape with three or more distinct byte values (writes + writer-head moves + source changes exceed 128). Theorem 2: if bytes written per byte of code executed is below one, a closed full replicator executes some address twice. Theorem 6: for any machine in which the pointer runs on into adjacent memory absent a control transfer and Theorem 2's bandwidth condition holds, a self-consistent literal-write instruction implies a straight-line (open) replicator the size of the literal, and its absence implies every replicator is loop-bearing. Lemma 7: $I(o; e | x) > 0$ implies imperfect fidelity across contexts; perfect fidelity implies $I = 0$. Proposition 3 (computer-assisted): the two-byte fixed points of the Z80 against the zero partner are exactly the five load–push words. Full statements and proofs in Supplementary Information.

BFF. cubff default semantics verified from source: two 64-byte tapes concatenated, pointer and heads at byte 0, ten instructions, every other byte a no-op, unmatched brackets halt, 2^{13} steps per encounter, 2^{17} programs, random ordered pairs each epoch, mutation 2^{-12} per byte per epoch. Switches: `ip_wrap` (pointer wraps modulo 128); `literal` (byte 0x50 = `P a b`: write `a` at `head1-1` and `b` at `head1-2`, then `head1 -= 2`, skip the operands); `nohalt` (unmatched brackets are no-ops). Our GPU kernel reproduces cubff's validation replicator and matches a CPU reference on randomised tests. 24

seeds each for the published and wrapping variants, 12 for each literal variant; events every 64 epochs; high-order entropy with brotli quality 2 as in ref. 5.

Searches. Two-byte census: all 65,536 words tiled to 16 bytes, culture test (32 partners) and partner test (256), plus the deterministic fixed-point test against the zero partner. BFF straight-line search: 16,447,860 programs (prefix ≤ 3 instructions + tiling unit of period 2–5 with a copy instruction), 2 random partners, wrapping pointer, alphabet prefilter then exact similarity, full culture test for survivors. Closed-replicator search: all 1,309,528 periodic tilings of $\{\mathbb{P}, [,], \text{no-op}\}$ up to period 10 under the literal, wrapping, no-halt rules, three partners each.

Pre-registration. Stages D, F and G and the BFF cells were pre-registered with predictions and kill criteria before launch; every amendment is dated in the change log. Misses reported: the heritable-fraction thresholds of G1(c) failed at $L = 50$; the syntactic loop criterion of G2 omitted hardware loops (6/20 and 3/20 syntactic against 19/20 and 8/20 behavioural); the mechanism first claimed for an open BFF replicator under a wrapping pointer was falsified by the harness test before any soup ran and the prediction restated with both readings; a held prediction for the non-wrapping literal cell was corrected by the counting argument before it was run.

Experimental stages. Stage A (630 runs, $L = 16$): seven ablations $\times$ budgets $\{32, 128, 512\} \times$ mutation $\{1/4, 1/16, 1/64\}$, ten seeds, 300,000 steps. Stage B (640): tape lengths 4–100 under two ablations and three budgets. Stage C (490): thirteen ablations at $L = 16$, seeds 101–110, with one-million-step nulls. Stage D (280, confirmatory): the $L = 9$ reversal with seeds 1001–1020. Stage E (1,230): the size axis under four control arms, budget and mutation sweeps at $L = 100$, variance and exception cells. Stage F (210, confirmatory): padded pair memories. Stage G (80, confirmatory): closure with seeds 2001–2020 at $L = 16$ and 50 (300,000 steps) and $L = 20$ and 64 (1,000,000 steps). BFF: 24 seeds each for the published and wrapping variants, 12 for each of the three literal variants. Post hoc analyses (closure, two-byte census, detectors, invasion, census dynamics, ring sweep) used recorded data and local GPU time only.

Compute, code and data. 3,560 Z80 runs and 84 BFF soups on Modal L40S GPUs plus local runs on an Apple GPU; $\approx$ \$400 at list price. The source code (simulator, assays, analysis and figure scripts), every condition file, every raw run summary, every generated table and the pre-registration documents with their dated change log are in the project repository (to be archived with a DOI on acceptance); each figure is produced from the generated tables by one script, and each number in the text is traceable to a named table.

Figure legends

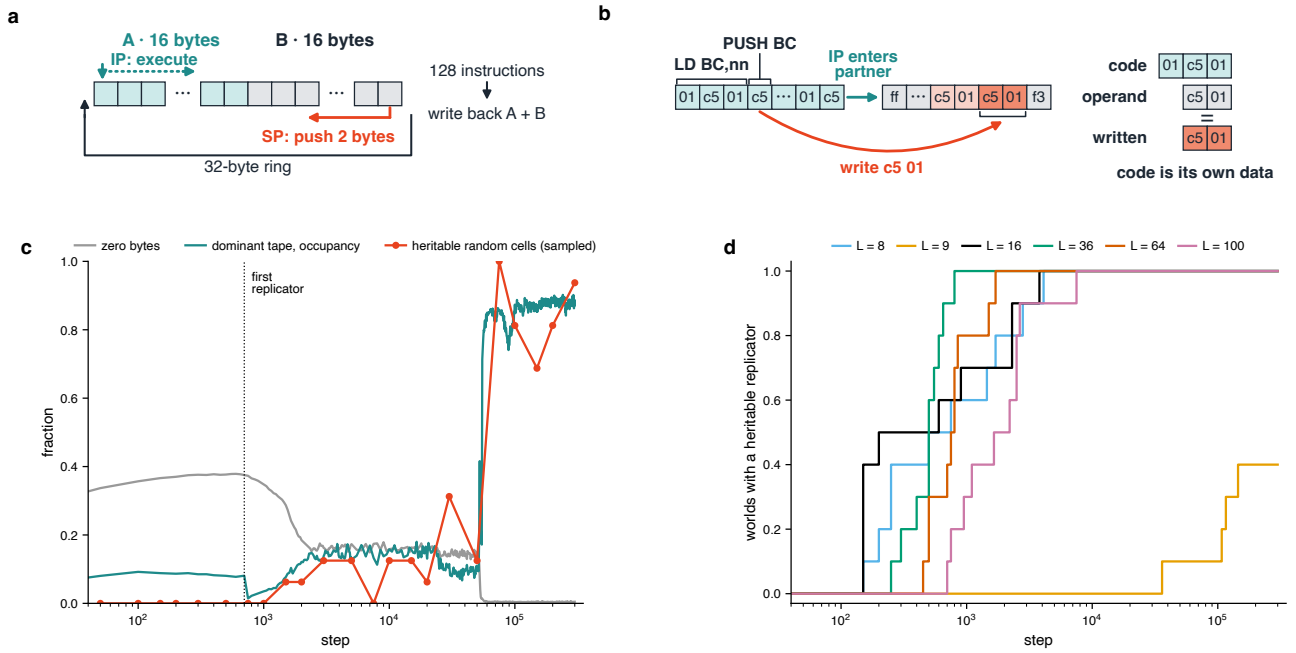


Fig. 1 | The soup and the order of events. **a**, An encounter: organism A (teal, 16 bytes) and partner B (grey, 16 bytes) share a 32-byte ring; the instruction pointer (IP, teal) starts at A's first byte and moves right, the stack pointer (SP, vermilion) starts at B's last byte and moves two bytes left with every push; after 128 instructions both halves are written back. Repeated cells are abbreviated. **b**, The first replicator, the word 01 c5 repeated: LD BC,nn loads its own next two bytes, c5 01, into BC, and PUSH BC writes exactly those two bytes at the stack pointer (vermilion; the lighter pair, the next push), so the partner fills from its far end with the organism's own word while the pointer runs on into the partner. Right, the instruction's bytes, its operand and what the push writes are the same word. **c**, One world ($L = 16$, seed 2001): fraction of zero bytes (the tar, grey), occupancy of the dominant tape (teal) and heritable fraction of 16 random cells at the sampled steps (vermilion, markers) against step; dotted line, first heritable replicator. **d**, Kaplan–Meier fraction of worlds with a heritable replicator by step for $L = 8$ –100 (Stage E, 10 worlds per length, 300,000 steps).

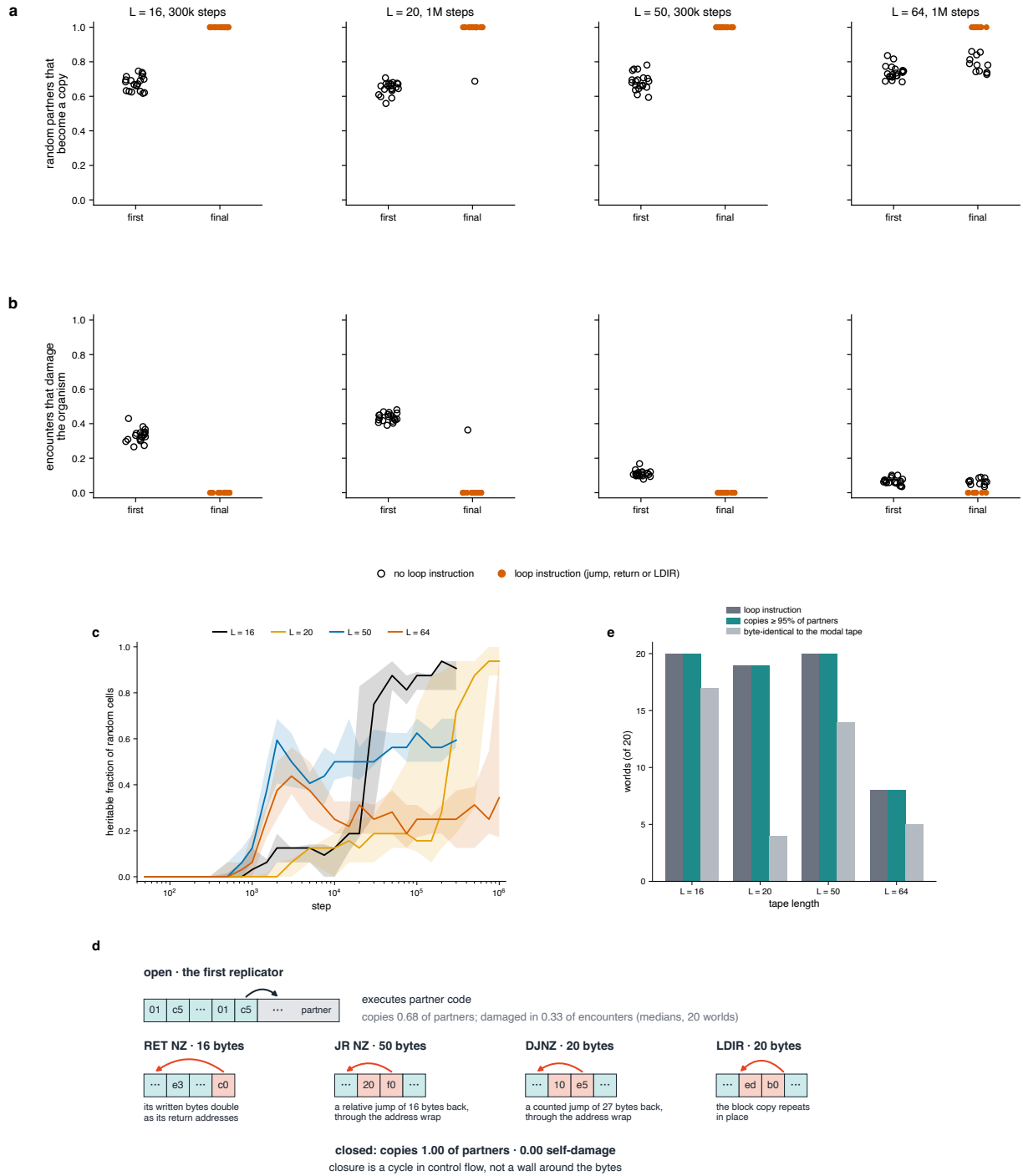


Fig. 2 | The first replicator is open, the successor closed. **a**, Fraction of 256 random partners that become a $\geq 75\%$ copy of the first heritable replicator and of the final dominant, one point per world (Stage G, 20 worlds per L; open circles, no loop instruction; filled, jump, return or LDIR). **b**, Fraction of encounters in which the organism loses $\geq 25\%$ of its bytes. **c**, Heritable fraction of 16 random cells against step (median and interquartile range over worlds). **d**, Control flow of the first replicator and of four closed successors, schematically (bytes abbreviated; the instruction that closes the cycle is shaded and its return drawn as a vermillion arc): the pusher's pointer runs on into the partner, whereas RET NZ returns to addresses formed from the organism's own written bytes, JR NZ and DJNZ jump back (through the 16-bit address wrap), and LDIR repeats in place. Medians of the partner test over Stage G worlds: the first replicator copies 0.68 of partners and is damaged in 0.33 of encounters; every closed successor copies 1.00 with 0.00 self-damage. Exact traces, Extended Data Fig. 2. **e**, Worlds (of 20) whose final dominant carries a loop instruction, copies $\geq 95\%$ of partners, and equals the modal final tape byte for byte.

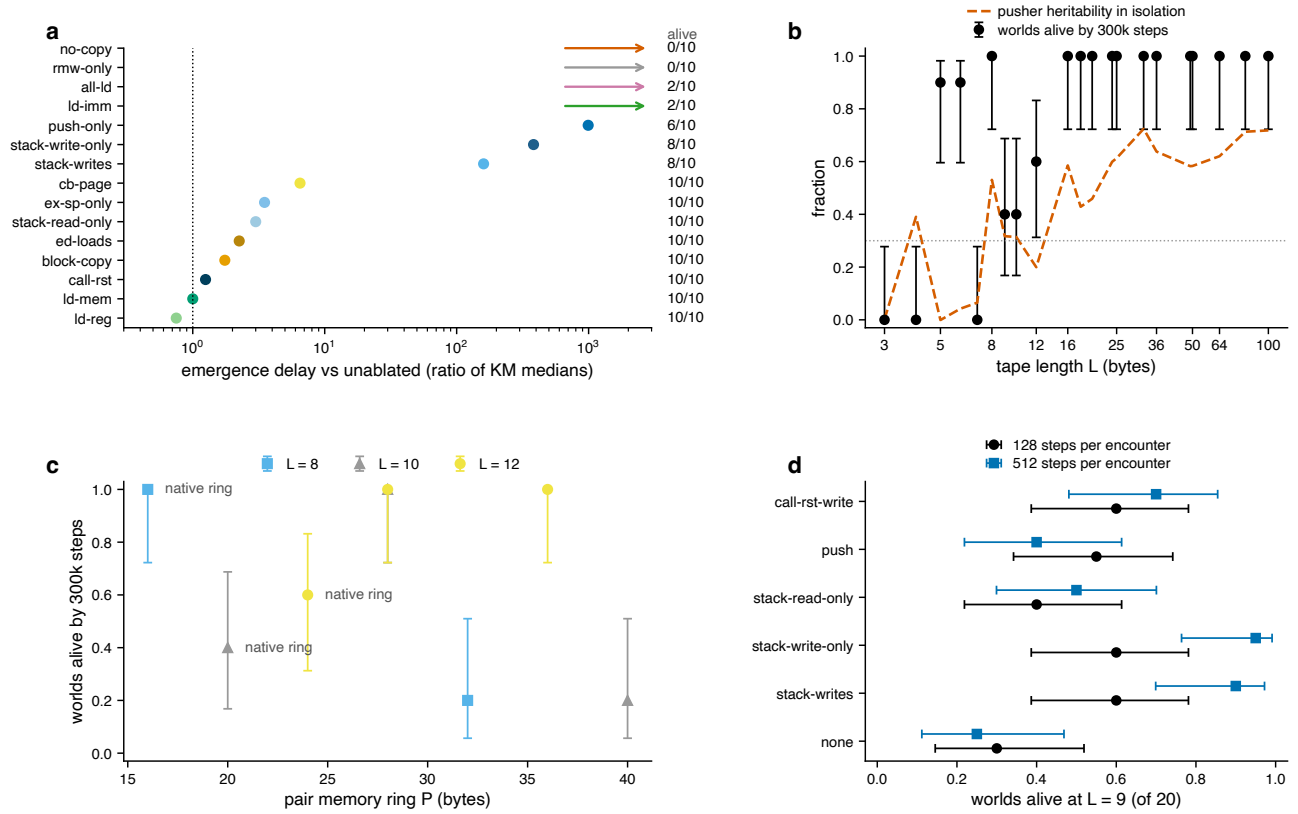


Fig. 3 | What the instruction set must provide. **a**, Delay of emergence relative to the unablated soup (ratio of Kaplan–Meier medians, 128 steps, mutation 1/16, seeds 101–110) for each ablation; arrows, no emergence; right margin, worlds alive by 300,000 steps. **b**, Worlds alive by 300,000 steps against tape length (Wilson 95% intervals) and the pusher’s heritability in isolation (dashed; dotted line, the 0.3 threshold). **c**, Worlds alive against pair-memory ring length P at L = 8, 10, 12 (Stage F; the native ring is 2L). **d**, Worlds alive at L = 9 (of 20) by ablation at 128 and 512 steps per encounter (Stage D).

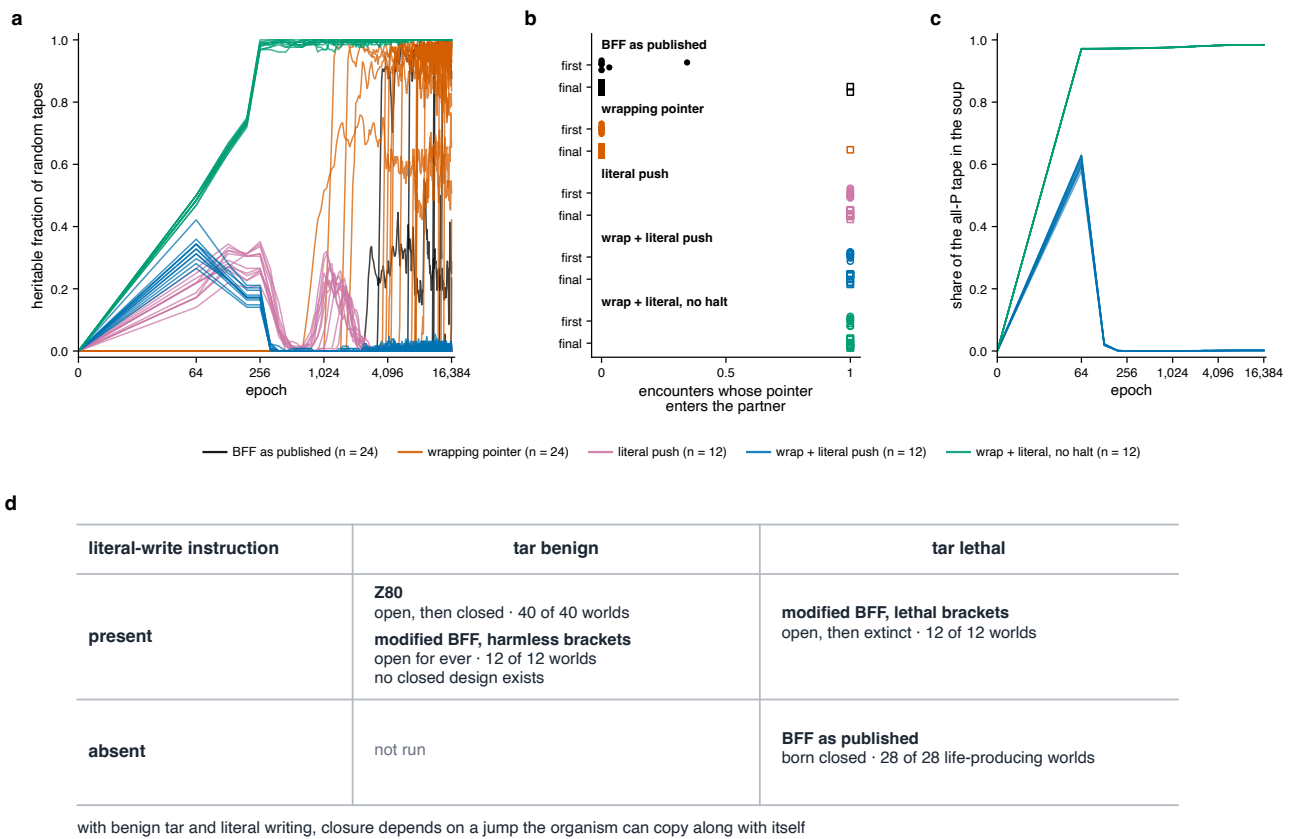


Fig. 4 | One instruction decides how life begins in BFF. **a**, Heritable fraction of 32 random tapes against epoch for every soup, by variant (as published, $n = 24$; wrapping pointer, $n = 24$; literal push, $n = 12$; wrap + literal push, $n = 12$; wrap + literal with unmatched brackets as no-ops, $n = 12$); the epoch axis is linear to 64 and logarithmic beyond, here and in **c**. **b**, Fraction of culture-test encounters in which the pointer enters the partner, for the first replicator (circles) and the final dominant (squares) of every life-producing soup, grouped by variant (open symbols, no loop instruction; filled, loop instruction). **c**, Share of the all-P class against epoch under wrap + literal (blue) and wrap + literal with no-op unmatched brackets (green), 12 worlds each (zero when the class is outside the ten largest): the same organism collapses when unmatched brackets halt the pointer and persists when they are no-ops. **d**, Classification of the substrates by two properties, a literal-write instruction (present or absent) and the lethality of the tar to an open replicator, with the observed order of events and world counts in each cell.

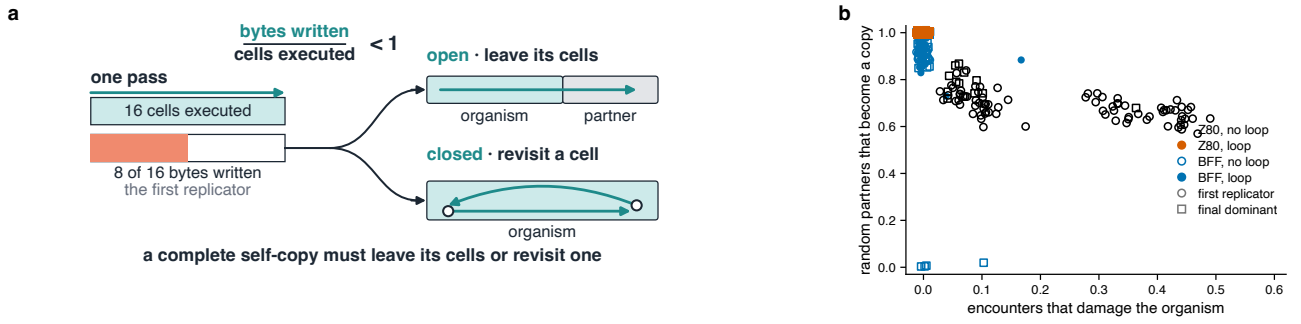


Fig. 5 | Closure as a cycle. **a**, Theorem 2 as a diagram. A program that writes fewer bytes than the cells it executes cannot copy itself in one pass: the first replicator ($L = 16$) executes its 16 cells and has written 8 of the 16 bytes a copy needs (executor trace), so copying requires more execution, which can only continue outside its cells (open) or revisit one of them (closed, a cycle). **b**, Copy success against self-damage for every measured first (circles) and final (squares) replicator: Z80 (black/vermilion) and BFF (blue); filled, loop-bearing. The closure-window model ($q \int n \, dt$) is in Extended Data.

Extended Data

Extended Data Fig. 1 | Emergence by tape length and ablation. Kaplan–Meier fraction of worlds with a heritable replicator against step for every tape length (4–100) under the four Stage E control arms (nominal; constant mutation per byte; constant mutation per cell; constant budget per byte) and under the stack-writer ablation; 10 worlds per cell, 300,000 steps. Source: `results/stageE/stage_e/size_arms.csv`, `results/stageE/assays.csv`.

Extended Data Fig. 2 | The replicator zoo. First heritable replicator and final dominant tape of every Stage G world (80 worlds), as hex and disassembly, grouped by tape length, with copy success and self-damage from the partner test. Source: `results/stageG/ZOO.md`, `results/stageG/stageG/stage_g_runs.csv`.

Extended Data Fig. 3 | Ring geometry. Worlds alive and median emergence step for every tested (tape length, ring length) pair of the ring sweep (52 cells) and Stage F; the pre-registered ring-arithmetic law (period divides the ring) against the observed law (period divides the pair length 2L); the stack-pointer/instruction-pointer race that produces the dead zone. Source: `results/ring_sweep/ring_sweep.csv`, `results/stageF/stage_f/rings.csv`, `results/stageF/stage_f/pusher_trace.csv`.

Extended Data Fig. 4 | Budget and mutation at L = 100. Heritable and faithful worlds against steps per encounter (32–2,048) under the pusher and LDIR regimes; the sub-clonal regime at 64 steps (heritable fraction of random cells against dominant-tape share); median period of the first replicator and final high-order entropy against per-byte mutation rate (the error-threshold curve). Source: `results/stageE/stage_e/budget_L100.csv`, `mutation_L100.csv`.

Extended Data Fig. 5 | Life detectors against the culture test. Area under the ROC curve for each recorded statistic (high-order entropy, non-unique tape fraction, bits per byte, species entropy, interaction-based statistics, dominant-tape share, quasispecies occupancy, motif share) against the heritable fraction of random cells, by tape length and by step; the inversion of high-order entropy under the zero flood. Source: `results/detectors/`.

Extended Data Fig. 6 | Tar prevents assembly. Census of LDIR-bearing tapes in the ten most common tapes before emergence at L = 9, with and without stack writers (20 worlds per arm): first entry, episodes, destruction rate and zero bytes written per interaction. Source: `results/stageD/census/`.

Extended Data Fig. 7 | Invasion assays. The pusher and the LDIR unit seeded into 1% of cells at 32 and 128 steps per encounter and two mutation rates: occupancy against step, 20 worlds per cell. Source: `results/invasion/invasion.csv`.

Extended Data Fig. 8 | Every two-byte word. Culture-test score and gen2 for all 65,536 two-byte words tiled to 16 bytes; the five fixed points against the zero partner; the 246 return-address smears. Source: `results/census2/`.

Extended Data Fig. 9 | Searches in BFF. The 16.4-million-program straight-line search under a wrapping pointer (copy fraction and offspring byte diversity of the 130 programs passing the byte-match criterion) and the 1.3-million-tiling closed-replicator search under the benign-tar rule (open replicators by period; no closed heritable tiling). Source: `results/bff_search/`, `results/bff_closed_search/`.

Extended Data Fig. 10 | Reproducibility and compute. Spread of emergence times across ten repeats of one seed against ten seeds in the Z80 soup (GPU scheduling nondeterminism); identity of epoch statistics between local Apple-GPU and Modal L40S runs of six BFF seeds over 16,384 epochs; GPU hours and cost per stage. Source: `results/stageE/stage_e/NUMBERS_E.md` (E6), `results/bff/reproducibility_local_vs_modal.csv`, `PLAN.md` change log.

Extended Data Table 1 | Final-tape classes per tape length (Stage G). Control-flow and block-repeat content, partner-copy fraction and self-damage of the final dominant, number of worlds per class. Source: `results/stageG/stageG/NUMBERS_G.md`.

Extended Data Table 2 | Ablation sets. Opcode families removed in each ablation, with opcode counts and example mnemonics. Source: `make_conds.ABLATIONS`, `src/lib/z80-opcodes.ts`.

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